Clustering

* Clustering is a M.L technique that groups similar data points points together into clusters based on their characterstics.

Types

1.) **Hard Clustering** :- Each data point strictly belongs to exactly one cluster, no overlap is allowed.

2.) **Soft Clustering** :- Soft clustering assigns each data point a probability or degree of membership to multiple clusters simultaneously.

Ex:- A datapoint may have a 70% membership in cluster 1 & 30% in cluster 2 reflecting uncertainty or overlap in group.

* Clustering methods can be classified on the basis of how they form clusters.

1.) Centroid-based Clustering

* This organizes data points around central prototypes called centroids, where each cluster is represented by mean.
* This is efficient for spherical & similarly sized clusters but sensitive to outliers & initialization.

Types

1.) K-means :- Iteratively assigns points to nearest centroid & re-calculates centroids to minimize intra-cluster variance.

2.) K-mediods :- Similar to K-means but uses actual data points as centers, robust to outliers.

2.) Density-based Clustering

* Clusters are differentiated as contiguous regions of high data density seperated by areas of lower density.

Algorithms:-

→ DBSCAN (Density-Based Spatial Clustering of Applications with noise).

→ OPTICS (Ordering Points To Identify Clustering Structure).

3) Connectivity-based Clustering

* Groupings of data by evaluating how data points are connected to their neighbors.

Algorithms

1) Agglomerative (Bottom-up):- Start with each point as a cluster.

2) Divisive (Top-down): Start with one cluster.

4) Distribution-based Clustering

* Assumes data is generated from a mixture of probability distributions.

Algorithm:-

1) Gaussian Mixture Model (GMM):- Fits data as a weighted mixture of gaussian distributions.

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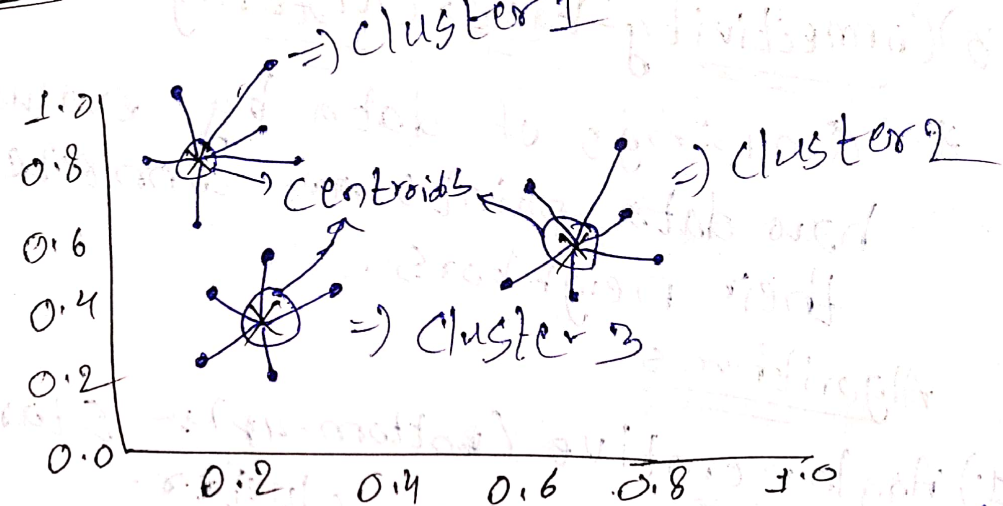
Fuzzy Clustering

* This allows each data point to belong to multiple clusters with varying degrees of membership.

K-means Clustering

Ex:-



* categorized the nearest data points to the nearest datapoint.
* For calculating similarity we use Euclidean distance.

Applications

1) Data Segmentation.
2.) Image Compression
3.) Anomaly Detection.
4) Document Clustering
5.) Organizing Large Datasets.

C-means Clustering Algorithm

14/10/25

* Unsupervised learning algorithm.
* In C-means (Fuzzy C-means), each data point can belong to multiple clusters with different degrees of membership.
* In K-means each data point belongs to exactly one cluster.

Applications.

→ Image segmentation.
→ Pattern recognition
→ Data compression.

Algorithm

1) Initialize:-

- Choose number of clusters
- Set the fuzziness parameter(m) (usually between 1.5 & 3).
- Randomly assign membership values b/w (0 & 1).

2) Computing cluster Centers:

$$V_j = \frac{\sum_{i=1}^{n} (u_{ij})^m \cdot x_j}{\sum_{i=1} (u_{ij})^m}$$

u_{ij} = membership degree of datapoint x_i in cluster j

m = fuzziness parameter.

3) Update Membership Values

$$u_{ij} = \frac{1}{\sum_{k=1} \left(\frac{||x_i - v_j||}{||x_i - v_k||} \right)^{\frac{2}{m-1}}}$$

4) Repeat steps 2-3 until the membership value stop changing significantly.

5.) Cluster centres & membership matrix

Ex:-

Data points: 1, 2, 5, 8.

No of clusters: $c = 2$

$V_1^{(0)} = 2.0$, $V_2^{(0)} = 7.0$

Iteration 1

$|x - v_1|$ & $|x - v_2|$

i) $x = 1$

$d1 = |1 - 2| = 1$

$d2 = |1 - 7| = 6$

(Cluster 1)

ii) $x = 2$

$d1 = |2 - 2| = 0$

$d2 = |2 - 7| = 5$

(Cluster 1)

iii) $x = 5$

$d1 = |5 - 2| = 3$

$d2 = |5 - 7| = 2$

(Cluster 2)

iv) $x = 8$

$d1 = |8 - 2| = 6$

$d2 = |8 - 7| = 1$

(Cluster 2)

* 1 & 2 are cluster 1
* 5 & 8 are cluster 2

$V_1^{(1)} = \text{mean}(1,2) = (1+2)/2 = 1.5$ } update
$V_2^{(1)} = \text{mean}(5,8) = (5+8)/2 = 6.5$ } centres

Iteration 2

$x = |1 - 1.5| = 0.5$ & $|1 - 6.5| = 5.5$ } Cluster 1
$x = |2 - 1.5| = 0.5$ & $|2 - 6.5| = 4.5$ }
$x = |5 - 1.5| = 3.5$ & $|5 - 6.5| = 1.5$ }
$x = |8 - 1.5| = 6.5$ & $|8 - 6.5| = 1.5$ } Cluster 2

15/10/25

Dimensionality Reduction

* This is the process of reducing the number of input variables in a dataset while retaining as much important information as possible.
* Simplify models.
* improve performance.
* Remove noise or redundancy.

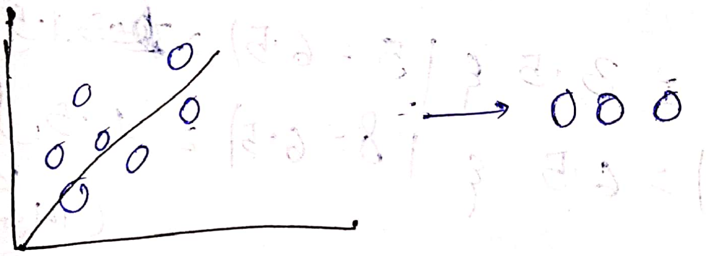
Uses:

1) Reduce overfitting.
2) Improve model accuracy.
3) Reduce Computational Time.
4) Visualization
5) Handle curse of dimensionality.

Types

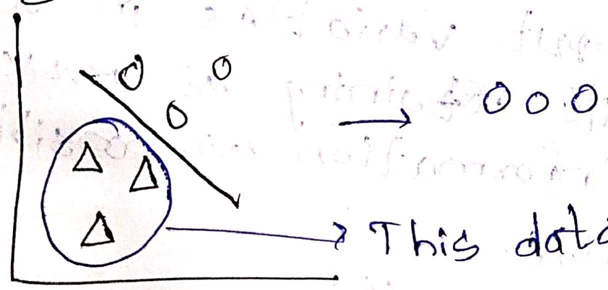
1) PCA
2) t-SNE
3) UMAP.
4) manifold Learning.
5) LDA.

PCA (Principal component Analysis).



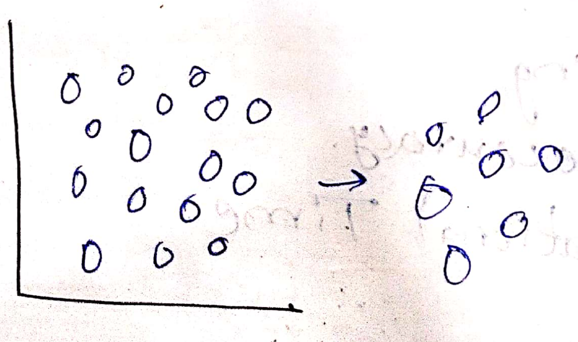
Linear

LDA (Linear Discriminant Analysis).

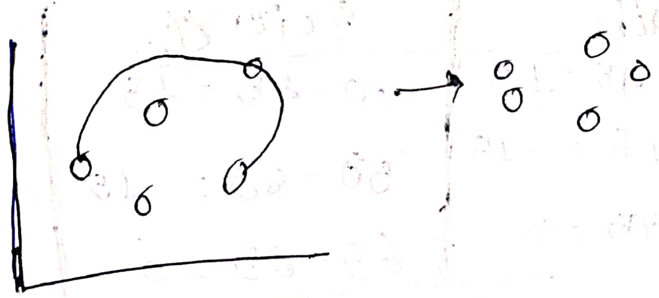


This data is not taken

t-SNE (t-Distributed Stochastic Neighbor Embedding). ⇒ Non Linear



UMAP (Uniform Manifold Approximation and Projection).



=) Non-Linear.

1) Feature selection

2) Feature Extraction

Applications

→ Image compression & recognition

→ Text & document classification

→ Genomics and bioinformatics.

→ Customer segmentation.

→ Data visualization (2D plots).

1) Principal Component Analysis

* Reduce correlated features into fewer uncorrelated components that capture the most variance.

Student	Math	Science
A	90	80
B	60	50
C	75	65

=) Reduce 2 features

↓

1 feature.

Step1:- standardize data.

Subject	Mean
Math	$(90+60+75)/3 = 75$
Science	$(80+50+65)/3 = 65$

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Step 2:-

Student	Math	Science
A	$90-75=15$	$80-65=15$
B	$60-75=-15$	$50-65=-15$
C	$75-75=0$	$65-65=0$

* Compute Covariance

$$Cov(X,Y) = \frac{1}{n-1} \Sigma (x_i - \bar{x})(y_i - \bar{y})$$

$$\boxed{Var = \frac{\Sigma (x_i - \bar{x})^2}{n}}$$

$$Cov = \begin{bmatrix} var(maths) & cov(maths, science) \\ cov(maths, science) & var(science) \end{bmatrix}$$

$$Var(maths) = Var(science) = \frac{15^2 + (-15)^2 + 0^2}{2}$$

$$= 225$$

$$Cov(maths, science) = \frac{225 + 225}{2} = 225.$$

Step 3:-

$$|Cov - \lambda I| = 0$$

$$\begin{vmatrix} 225-\lambda & 225 \\ 225 & 225-\lambda \end{vmatrix} = 0$$

$$(225-\lambda)^2 - 225^2 = 0$$

$$\lambda_1 = 450, \lambda_2 = 0.$$

Step 4:- Project data on Principal Component

$$PC_1 = \frac{1}{\sqrt{2}}(maths + science)$$

Student	Math	Science	PC1
A	15	15	$(15+15)/\sqrt{2} = 21.2$
B	-15	-15	$(-30)/\sqrt{2} = -21.21$
C	0	0	0

Real life uses

→ Image compression

→ Feature reduction before training ML models.

→ Customer behaviour

LDA (Linear Discriminant Analysis)

* Reduce feature while maximizing class seperation.

Ex:-

Fruit	Weight	Sweetness	Class
A	1.0	1.0	Apple
B	2.0	1.5	Apple
C	2.0	3.0	Orange.
D	3.0	3.5	Orange

Step 1:- Compute class means.

*Apple mean : (Weight = 1.5, Sweetness = 1.25)

*Orange mean : (Weight = 2.5, Sweetness = 3.25)

A → (1, 1) − (1.5, 1.25) = (-0.5, 0.25)

B → (2, 1.5) − (1.5, 1.25) = (0.5, 0.25)

C → (-0.5, -0.25) −

D → (0.5, 0.25)

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Step 3:-

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

$$\mu_1 - \mu_2 = (1.5 - 2.5,\ 1.25 - 3.25) = (-1, -2)$$

$$S_B = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Step 4:-

Compute $S_W^{-1} S_B$

Step 5:-

Fruit	New 1D Value
A	$1 = 0.45 + 1 \times 0.89 = 1.34$
B	$2 = 0.45 + 1.5 \times 0.89 = 2.22$
C	$2 = 0.45 + 3 \times 0.89 = 3.63$
D	$3 = 0.45 + 3.5 \times 0.89 = 4.46$

MANIFOLD LEARNING

* High dimensional data lies on or near a low-dimensional manifold embedded in the higher-dimensional space. The data ~~lea~~ might have many features
* non-linear dimensionality reduction.

Applications

→ Bioinformatics
→ NLP
→ Computer Vision
→ Anomaly detection.

T-SNE (T - Distributed Stochastic neighbor Embedding).

* focuses on preserving local structure in data.
* Converts distances into probabilities of similarity.
* Great visualization of local patterns & clusters.
* Reveals hidden structure in data.

UMAP (UNIFORM MANIFOLD APPROXIMATION AND PROJECTION)

* Preserves ~~and pre pr~~ both local & global structure.
* Visualization & pre-processing step for ML
* Good for large datasets.

Anomaly Detection

* Anamalies also known as outliers are data points that deviate significantly from the expected behaviour.

Types

1) Point Anomalies

* Represents individual data points that are statistically unusual compared to the rest of the data.

2) Contextual Anomalies

* Occurs in time series data.
* Anomalies depend on the specific context or environment surrounding them.

3) Collective Anomalies

* Groups of related data points exhibiting abnormal behavior collectively.

Isolation Forests

* Isolation forest is an unsupervised anomaly detection algorithm which is useful high dimensional data.
* Isolates anomalies.

Working

→ Building Isolation Trees:-
Trees are created, typically hundreds or even thousands of them. These aim to isolate individual data points.

→ Splitting on random features:
* Isolation trees introduce randomness at each node of the tree.
* This randomness helps ensure that anomalies.

→ Isolating Data Points:–
The data points are then directed down the branches of the isolation tree based on their feature values.
* If a data point < split value (Left branch)
* If a datapoint > split value (Right branch).

→ Anomaly Score:–
* By using the path length of a data point through an isolation tree, anomalies can be isolated easily.

→ Anomaly Score Calculation:– For each tree the path length is required to isolate the data point.

→ Identifying Anomalies:– Data points with shorter average path lengths are considered more likely to be anomalies.

Local Outlier Factor (LOF)
* Used for unsupervised outlier detection.
* Local density is calculated by estimating distances between data points that are neighbours i.e; K-nearest neighbors.

1) K - Distance & K - Nearest Neighbors.

2) Reachability Distance

* The max distance b/wn two points & the K - distance of the point.

$$rd_k(A,B) = \max\{k\text{-distance}(B), d(A,B)\}$$

$$LOF_k(A) = \frac{\sum_{B \in N_k(A)} LRD_k(B)}{|N_k(A)| \cdot LRD_k(A)}$$

→ A = data point

K = No of neighbors.

$N_k(A)$ = set of k-nearest neighbors of A.

$|N_k(A)|$ = no of neighbors in the set.

$LRD_k(A)$ = Local Reachability Density.

$$LRD_k(A) := \frac{1}{\left(\frac{\sum_{B \in N_k(A)} \text{reachability-distance}_k(A,B)}{|N_k(A)|}\right)}$$

LOF ~ 1 = similar data point.

LOF < 1 = data point which is inside the density cluster.

LOF > 1 = outlier.

Core Principles of AI Ethics

→ Transparency
→ Accountability.
→ Fairness
→ Privacy & Security.
→ Human-Centric Design.
→ Fairness

Sources of Bias in AI

→ Biased Training Data
* Historical data may reflect societal biases.

→ Unrepresentative Samples
* Lack of diversity in data can cause poor performance on minority or less frequent groups.

→ Human Prejudices in Labeling
* Annotators may unconsciously inject their own biases during data labeling

→ Algorithmic Design & Feature Selection
* Choices in algorithms or features may reinforce bias.

Pre-processing:- Modify training data before model building.

Techniques:- 1) Re-sampling.
2) Re-weighting

In-processing:- Modify the learning algorithm to enforce fairness.

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Post-processing

* Adjust model predictions after training.

→ Threshold optimization

→ Calibration.

Responsible AI:-

The practical implementation of AI ethics, focusing on building, deploying & managing AI systems in a way that is fair, transparent, accountable.

Challenges in implementing responsible AI

1) Data Bias
2) Lack of Transparency.
3) Defining Accountability.
4) Regulatory Fragmentation
5) Balancing Innovation & Ethics.
6) Technological & Cultural Resistance.